

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: DVR
Manufacturer: Blackmagic Design
Firmware Version: N/A
Model(s): HyperDeck Studio, HyperDeck Studio Pro, HyperDeck Studio 12G, HyperDeck Studio Mini

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_3_2_0	11/29/18	Fixed Transport Information match string for Ethernet control.
1_3_1_0	6/21/2018	Removed status for Play Single Clip command for Ethernet control. Fixed Transport Information status for Ethernet control.
1_2_0_0	10/2/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- For Ethernet commands, call Update function for ClipCount and TransportInformation before any Set command, as the command string gets built with the status of those two commands.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='HyperDeck Studio')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='HyperDeck Studio')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 9993, Model='HyperDeck Studio')</code>

Set Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Jog	Value 'Fwd 1' 'Rev 2'	Value 'Fwd 2'	Value 'Rev 1'
# Jog example InterfaceName.Set('Jog', 'Fwd 1')			
Command Preview	Value None		
# Preview example InterfaceName.Set('Preview', None)			
Command Review	Value None		
# Review example InterfaceName.Set('Review', None)			
Command Shuttle	Value 'Fwd 1' 'Rev 2'	Value 'Fwd 2'	Value 'Rev 1'
# Shuttle example InterfaceName.Set('Shuttle', 'Fwd 1')			
Command Standby	Value 'On'	Value 'Off'	
# Standby example InterfaceName.Set('Standby', 'On')			
Command Transport	Value 'Play' 'Next' 'Reverse' 'Start' 'Fast Fwd'	Value 'Stop' 'Previous' 'Fast Forward' 'End' 'Rew'	Value 'Record' 'Forward' 'Fast Reverse' 'Eject'
# Transport example InterfaceName.Set('Transport', 'Play')			

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus, and VideoFormat do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Standby' 'Rew'	Value 'Stop' 'Fast Fwd'	Value 'Play' 'Record'
# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			

Set Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AudioInput ¹	Value 'Embedded'	Value 'XLR'	Value 'RCA'
# AudioInput example InterfaceName.Set('AudioInput', 'Embedded')			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Jog	Value 'Fwd 1' 'Rev 2'	Value 'Fwd 2'	Value 'Rev 1'
# Jog example InterfaceName.Set('Jog', 'Fwd 1')			
Command Override	Value 'On'	Value 'Off'	
# Override example InterfaceName.Set('Override', 'On')			
Command PlayLoop	Value 'On'	Value 'Off'	
# PlayLoop example InterfaceName.Set('PlayLoop', 'On')			
Command PlaySingleClip	Value 'On'	Value 'Off'	
# PlaySingleClip example InterfaceName.Set('PlaySingleClip', 'On')			
Command PlaySpeed	Value -1600 to 1600 in steps of 1		
# PlaySpeed example InterfaceName.Set('PlaySpeed', 1600)			
Command PreviewMode	Value 'On'	Value 'Off'	
# PreviewMode example InterfaceName.Set('PreviewMode', 'On')			
Command SlotID	Value '1'	Value '2'	

Global Scripter Module Communication Sheet

# SlotID example InterfaceName.Set('SlotID', '1')			
Command TimecodeGoTo	Value 'Timecode'	Value 'Display Timecode'	
Qualifier Key 'Action'	Qualifier Value 'Go to Specified Timecode'	Qualifier Value 'Move Forward'	Qualifier Value 'Move Backward'
# TimecodeGoTo example InterfaceName.Set('TimecodeGoTo', 'Timecode', {'Action': 'Go to Specified Timecode'})			
Command TimecodeGoToClip	Value 'Timecode Start'	Value 'Timecode Duration'	
Qualifier Key 'Action'	Qualifier Value 'Go to Specified Timecode'	Qualifier Value 'Move Forward'	Qualifier Value 'Move Backward'
Qualifier Key 'Clip ID'	Qualifier Value '1' – '255'		
# TimecodeGoToClip example InterfaceName.Set('TimecodeGoToClip', 'Timecode Start', {'Action': 'Go to Specified Timecode', 'Clip ID': '1'})			
Command TimecodeMemory	Value '1' – '10'		
Qualifier Key 'Action'	Qualifier Value 'Save'	Qualifier Value 'Recall'	
# TimecodeMemory example InterfaceName.Set('TimecodeMemory', '1', {'Action': 'Save'})			
Command Transport	Value 'Play' 'Next' 'Reverse' 'Start' 'Fast Fwd'	Value 'Stop' 'Previous' 'Fast Forward' 'End' 'Rew'	Value 'Record' 'Forward' 'Fast Reverse' 'Eject'
# Transport example InterfaceName.Set('Transport', 'Play')			
Command VideoFormat	Value 'NTSC' 'PALp' '720p60' '1080p25' '1080i50' '4Kp23976' '4Kp2997' '4Kp5994'	Value 'PAL' '720p50' '1080p23976' '1080p2997' '1080i5994' '4Kp24' '4Kp30' '4Kp60'	Value 'NTSCp' '720p5994' '1080p24' '1080p30' '1080i60' '4Kp25' '4Kp50'

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# VideoFormat example InterfaceName.Set('VideoFormat', 'NTSC')			
Command VideoInput ²	Value 'SDI'	Value 'HDMI'	Value 'Component'
# VideoInput example InterfaceName.Set('VideoInput', 'SDI')			

1 Only available for HyperDeck Studio Pro.

2 Not available for HyperDeck Studio Mini.

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus, and VideoFormat do not support the Update function. ConnectionStatus is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	Value
AudioInput	'Embedded'	'XLR'	'RCA'
# AudioInput examples InterfaceName.Update('AudioInput') Value = InterfaceName.ReadStatus('AudioInput') InterfaceName.SubscribeStatus('AudioInput', None, FeedbackHandler)			
Command	Value		
ClipCount	Decimal		
# ClipCount examples InterfaceName.Update('ClipCount') Value = InterfaceName.ReadStatus('ClipCount') InterfaceName.SubscribeStatus('ClipCount', None, FeedbackHandler)			
Command	Value		
ClipName	'String'		
Qualifier Key	Qualifier Value		
'Clip ID'	'1' – '255'		
# ClipName examples InterfaceName.Update('ClipName', {'Clip ID': '1'}) Value = InterfaceName.ReadStatus('ClipName', {'Clip ID': '1'}) InterfaceName.SubscribeStatus('ClipName', None, FeedbackHandler)			
Command	Value		
ClipTimecodeDuration	'String'		
Qualifier Key	Qualifier Value		
'Clip ID'	'1' – '255'		
# ClipTimecodeDuration examples InterfaceName.Update('ClipTimecodeDuration', {'Clip ID': '1'}) Value = InterfaceName.ReadStatus('ClipTimecodeDuration', {'Clip ID': '1'}) InterfaceName.SubscribeStatus('ClipTimecodeDuration', None, FeedbackHandler)			

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Command ClipTimecodeStart	Value 'String'		
Qualifier Key 'Clip ID'	Qualifier Value '1' – '255'		
# ClipTimecodeStart examples InterfaceName.Update('ClipTimecodeStart', {'Clip ID': '1'}) Value = InterfaceName.ReadStatus('ClipTimecodeStart', {'Clip ID': '1'}) InterfaceName.SubscribeStatus('ClipTimecodeStart', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command ErrorStatus	Value 'Syntax Error' 'Unsupported' 'Disk Error' 'No Input' 'Invalid State'	Value 'Unsupported Parameter' 'Disk Full' 'Timeline Empty' 'Remote Control Disabled'	Value 'Invalid Value' 'No Disk' 'Out Of Range' 'Connection Rejected'
# ErrorStatus examples Value = InterfaceName.ReadStatus('ErrorStatus') InterfaceName.SubscribeStatus('ErrorStatus', None, FeedbackHandler)			
Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command Override	Value 'On'	Value 'Off'	
# Override examples InterfaceName.Update('Override') Value = InterfaceName.ReadStatus('Override') InterfaceName.SubscribeStatus('Override', None, FeedbackHandler)			
Command PlayLoop	Value 'On'	Value 'Off'	
# PlayLoop examples Value = InterfaceName.ReadStatus('PlayLoop') InterfaceName.SubscribeStatus('PlayLoop', None, FeedbackHandler)			
Command PlaySpeed	Value -1600 to 1600 in steps of 1		
# PlaySpeed examples Value = InterfaceName.ReadStatus('PlaySpeed') InterfaceName.SubscribeStatus('PlaySpeed', None, FeedbackHandler)			

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Command SlotID	Value '1' – '2'	Value None	
# SlotID examples Value = InterfaceName.ReadStatus('SlotID') InterfaceName.SubscribeStatus('SlotID', None, FeedbackHandler)			
Command TimecodeDisplayStatus	Value 'String'		
# TimecodeDisplayStatus examples InterfaceName.Update('TimecodeDisplayStatus') Value = InterfaceName.ReadStatus('TimecodeDisplayStatus') InterfaceName.SubscribeStatus('TimecodeDisplayStatus', None, FeedbackHandler)			
Command TimecodeMemoryStatus	Value 'String'		
Qualifier Key 'Number'	Qualifier Value '1' – '10'		
# TimecodeMemoryStatus examples InterfaceName.Update('TimecodeMemoryStatus', {'Number': '1'}) Value = InterfaceName.ReadStatus('TimecodeMemoryStatus', {'Number': '1'}) InterfaceName.SubscribeStatus('TimecodeMemoryStatus', None, FeedbackHandler)			
Command TimecodeStatus	Value 'String'		
# TimecodeStatus examples InterfaceName.Update('TimecodeStatus') Value = InterfaceName.ReadStatus('TimecodeStatus') InterfaceName.SubscribeStatus('TimecodeStatus', None, FeedbackHandler)			
Command TransportInformation	Value 'Playing' 'Forward' 'Rewind'	Value 'Stopped' 'Jog' 'Preview'	Value 'Record' 'Shuttle'
# TransportInformation examples InterfaceName.Update('TransportInformation') Value = InterfaceName.ReadStatus('TransportInformation') InterfaceName.SubscribeStatus('TransportInformation', None, FeedbackHandler)			
Command VideoFormat	Value 'NTSC' 'PALp' '720p60' '1080p25' '1080i50' '4Kp23976' ¹ '4Kp2997' ¹ '4Kp5994' ²	Value 'PAL' '720p50' '1080p23976' '1080p2997' '1080i5994' '4Kp24' ¹ '4Kp30' ¹ '4Kp60' ²	Value 'NTSCp' '720p5994' '1080p24' '1080p30' '1080i60' '4Kp25' ¹ '4Kp50' ²

```
# VideoFormat examples
Value = InterfaceName.ReadStatus('VideoFormat')
InterfaceName.SubscribeStatus('VideoFormat', None, FeedbackHandler)
```

Command	Value	Value	Value
VideoInput ³	'SDI'	'HDMI'	'Component'

```
# VideoInput examples
InterfaceName.Update('VideoInput')
Value = InterfaceName.ReadStatus('VideoInput')
InterfaceName.SubscribeStatus('VideoInput', None, FeedbackHandler)
```

¹ Only available for HyperDeck Studio Pro.

² Only available for HyperDeck Studio 12G.

³ Not available for HyperDeck Studio Mini.

Cable and Adapter Requirements

Captive Screw connector to Male RS-422 DB9 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-422

Baud Rate: 38400

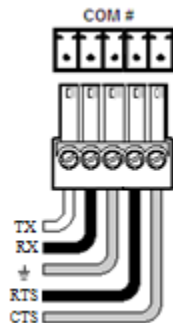
Data Bits: 8

Parity: Odd

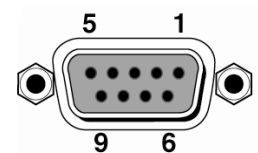
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
Tx- (Tx)	→	2	Rx-
Rx- (Rx)	←	8	Tx-
GND	→	9	GND
Tx+ (RTS)	→	7	Rx+
Rx+ (CTS)	←	3	Tx+



**RS422 DB9, Female
Video Recorder**

Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	9993
Logon Credentials Supported:	No
Multi-Connection Capabilities:	No
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands (Serial)

Freeze Off	j\x8A
Freeze On	k\x8B
Jog Fwd 1	!\x112
Jog Fwd 2	"\x113
Jog Rev 1	!!B
Jog Rev 2	"!C
Preview None	@`
Review None	Aa
Shuttle Fwd 1	!\x134
Shuttle Fwd 2	"\x135
Shuttle Rev 1	!#D
Shuttle Rev 2	"#E
Standby Off	\x04\$
Standby On	\x05%
Transport Eject	\x0F/
Transport Fast Fwd	\x100
Transport Play	\x01!
Transport Record	\x02"
Transport Rew	@
Transport Stop	\x00

Appendix B. Update Commands (Serial)

Device Status	p \x90
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Appendix C. Set Commands (Ethernet)

Audio Input Embedded	configuration: audio input: embedded\x0D\x0A
Audio Input RCA	configuration: audio input: RCA\x0D\x0A
Audio Input XLR	configuration: audio input: XLR\x0D\x0A
Executive Mode Off	remote: enable: true\x0D\x0A
Executive Mode On	remote: enable: false\x0D\x0A
Override Off	remote: override: false\x0D\x0A
Override On	remote: override: true\x0D\x0A
Play Loop Off	play: loop: false\x0D\x0A
Play Loop On	play: loop: true\x0D\x0A
Play Single Clip Off	play: single clip: false\x0D\x0A
Play Single Clip On	play: single clip: true\x0D\x0A
Play Speed 1600	play: speed: 1600\x0D\x0A
Play Speed -1600	play: speed: -1600\x0D\x0A
Preview Mode Off	preview: enable: false\x0D\x0A
Preview Mode On	preview: enable: true\x0D\x0A
Slot ID 1	slot select: slot id: 1\x0D\x0A
Slot ID 2	slot select: slot id: 2\x0D\x0A
Transport End	goto: clip: end\x0D\x0A
Transport Fast Forward	goto: timecode: +00:00:00:05\x0D\x0A
Transport Fast Reverse	goto: timecode: -00:00:00:05\x0D\x0A
Transport Forward	goto: timecode: +00:00:00:01\x0D\x0A
Transport Next	goto: clip id: +1\x0D\x0A
Transport Play	play\x0D\x0A
Transport Previous	goto: clip id: -1\x0D\x0A
Transport Record	record\x0D\x0A
Transport Reverse	goto: timecode: -00:00:00:01\x0D\x0A
Transport Start	goto: clip: start\x0D\x0A
Transport Stop	stop\x0D\x0A
Video Format 1080i50	slot select: video format: 1080i50\x0D\x0A
Video Format 1080i5994	slot select: video format: 1080i5994\x0D\x0A
Video Format 1080i60	slot select: video format: 1080i60\x0D\x0A
Video Format 1080p23976	slot select: video format: 1080p23976\x0D\x0A
Video Format 1080p24	slot select: video format: 1080p24\x0D\x0A
Video Format 1080p25	slot select: video format: 1080p25\x0D\x0A
Video Format 1080p2997	slot select: video format: 1080p2997\x0D\x0A
Video Format 1080p30	slot select: video format: 1080p30\x0D\x0A

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Video Format 4Kp23976	slot select: video format: 4Kp23976\x0D\x0A
Video Format 4Kp24	slot select: video format: 4Kp24\x0D\x0A
Video Format 4Kp25	slot select: video format: 4Kp25\x0D\x0A
Video Format 4Kp2997	slot select: video format: 4Kp2997\x0D\x0A
Video Format 4Kp30	slot select: video format: 4Kp30\x0D\x0A
Video Format 4Kp50	slot select: video format: 4Kp50\x0D\x0A
Video Format 4Kp5994	slot select: video format: 4Kp5994\x0D\x0A
Video Format 4Kp60	slot select: video format: 4Kp60\x0D\x0A
Video Format 720p50	slot select: video format: 720p50\x0D\x0A
Video Format 720p5994	slot select: video format: 720p5994\x0D\x0A
Video Format 720p60	slot select: video format: 720p60\x0D\x0A
Video Format NTSC	slot select: video format: NTSC\x0D\x0A
Video Format NTSCp	slot select: video format: NTSCp\x0D\x0A
Video Format PAL	slot select: video format: PAL\x0D\x0A
Video Format PALp	slot select: video format: PALp\x0D\x0A
Video Input Component	configuration: video input: component\x0D\x0A
Video Input HDMI	configuration: video input: HDMI\x0D\x0A
Video Input SDI	configuration: video input: SDI\x0D\x0A

Appendix D. Update Commands (Ethernet)

Audio Input	configuration\x0D\x0A
Clip Count	clips get\x0D\x0A
Executive Mode	remote\x0D\x0A
Override	remote\x0D\x0A
Transport Information	transport info\x0D\x0A
Video Input	configuration\x0D\x0A